In [50]:
```
if True:
   print('Data Science')
```

```
Data Science
```

In [51]:
```
if True:
   print('Data Science')
else:
    print('thank you')
```

```
Data Science
```

In [52]:
```
# check if number is even or odd
x=4
r= x % 2
if r==0:
    print('Even number')
```

```
Even number
```

In [53]:
```
x=5
r= x % 2
if r==0:
    print('Even number')
```

In [54]:
```
x=5
r= x % 2
if r==0:
    print('Even number')

if r==1:
    print('Odd number')
# we should not write multiple Ifs moaximum, we need to use mostly else ifs as well
```

```
Odd number
```

In [55]:
```
x=7
r= x % 2
if r==0:
    print('Even number')

else:
    print('Odd number')
```

```
Odd number
```

In [56]:
```
# NESTED IF
```

In [57]:
```
x = 3
r = x%2

if r==0:
    print('Even number')
    if x > 5:
        print('Greater value')
```

```
else:
    print('Odd Number')
```

```
Odd Number
```

In [58]:
```
x = 8
r = x%2

if r==0:
    print('Even number')
    if x > 5:
        print('Greater value')

else:
    print('Odd Number')
```

```
Even number
Greater value
```

x = 4 r = x%2

if r==0: print('Even number') if x > 5: print('Greater value') else: print('Less than value')

else: print('Odd Number')

IF-ELIF-ELSE

In [59]:
```
x=1

if x==1:
    print('one')
if x==2:
    print('two')
if x==3:
    print('three')
if x==4:
    print('four')
if x==5:
    print('five')
```

```
one
```

In [60]:
```
x=2

if x==1:
    print('one')
elif x==2:
    print('two')
elif x==3:
    print('three')
elif x==4:
    print('four')
```

```
elif x==5:
    print('five')
```

```
two
```

In [61]:
```
x=7

if x==1:
    print('one')
elif x==2:
    print('two')
elif x==3:
    print('three')
elif x==4:
    print('four')
elif x==5:
    print('five')
else:
    print('Number Not Found')
```

```
Number Not Found
```

In [62]:
```
# LOOPS ARE 2 TYPES:
# WHILE LOOP
#FOR LOOP
```

In [63]:
```
i = 1

while i <= 5:
    print('Data Science')
    i = i + 1
```

```
Data Science
Data Science
Data Science
Data Science
Data Science
```

In [64]:
```
i = 5

while i >= 1:
    print('Data Science:', i)
    i = i - 1
```

```
Data Science: 5
Data Science: 4
Data Science: 3
Data Science: 2
Data Science: 1
```

In [65]:
```
i = 1

while i<=5:
    print('Data Science:',i)
    j=1
    while j<=4:
        print('technology:',j)
```

```
        j=j+1
    i = i+1
    print()
# its a nested while
```

```
Data Science: 1
technology: 1
technology: 2
technology: 3
technology: 4

Data Science: 2
technology: 1
technology: 2
technology: 3
technology: 4

Data Science: 3
technology: 1
technology: 2
technology: 3
technology: 4

Data Science: 4
technology: 1
technology: 2
technology: 3
technology: 4

Data Science: 5
technology: 1
technology: 2
technology: 3
technology: 4
```

In [66]:

```
i = 1

while i<=5:
    print('Data Science:',end="")
    j=1
    while j<=4:
        print('technology:',end="")
        j=j+1
    i = i+1
    print()
# end="" parameter stops createing a new line and saves space
```

```
Data Science:technology:technology:technology:technology:
Data Science:technology:technology:technology:technology:
Data Science:technology:technology:technology:technology:
Data Science:technology:technology:technology:technology:
Data Science:technology:technology:technology:technology:
```

In [67]:

```
i = 1

while i<=2:
```

```
    j=0
    while j<=2:
        print(i*j,end="")
        j+=1
    print()
    i +=1
```

```
012
024
```

In [68]:
```
i = 1

while i<=4:
    j=0
    while j<=3:
        print(i*j,end="")
        j+=1
    print()
    i +=1
```

```
0123
0246
0369
04812
```

for loop

In [69]:
```
name = 'nit'  #i means iterations, it checks the values

for i in name:
    print(i)

#when data is string u can do while loop now
```

```
n
i
t
```

In [71]:
```
for i in range(5):
    print(i)
```

```
0
1
2
3
4
```

In [72]:
```
for i in range(1,10,3):
    print(i)
```

```
1
4
7
```

In [74]:
```
#print the value which is divisable by 5
for i in range(1,21):
```

```
    print(i)
```

```
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
```

In [75]:
```
#print the value which is divisable by 5
for i in range(1,21):
    if i%5==0:
        print(i)
```

```
5
10
15
20
```

In [77]:
```
for i in range(1,21):
    if i%5!=0:
        print(i)
```

```
1
2
3
4
6
7
8
9
11
12
13
14
16
17
18
19
```

BREAK CONTINUE PASS

In [ ]:
```
# conditional statement is used only if use case have 1 binary condition
#it uses nested if if there are more than 1 conditions.
#WHILE LOOP: We use while if use case has no binary conditions(repititive task)
#FOR LOOP: if we have some data sequences like string, list, tuple, dict, set we us
# We use PASS in method overriding ,polymorphis, abstract method
```

In [79]:
```
n=5
for i in range(2,n):
    if n%i==0:
        print("Not Prime")
        break
    else:
        print("Prime")
```

```
Prime
Prime
Prime
```

In [80]:
```
for i in range(1,11):
    if i==6:
        break
    print(i)
    # this condition will break the loop and gives only before condition
```

```
1
2
3
4
5
```

In [81]:
```
for i in range(1,11):
    if i==6:
        continue
    print(i)
#it continues the loop and skips the 6 value
```

```
1
2
3
4
5
7
8
9
10
```

In [82]:
```
for i in range(1,11):
    if i==6:
        continue
    print("hello:",i)
```

```
hello: 1
hello: 2
hello: 3
hello: 4
hello: 5
hello: 7
hello: 8
hello: 9
hello: 10
```

In [88]:
```
for i in range(1,11):
    pass

#here we get error since we didnt print any values but when we give pass it skips t
```

In [89]:
```
for i in range(1,21):
    if i%3==0:
        continue
    print(i)
print('end')

#it skips all 3 multiples
```

```
1
2
4
5
7
8
10
11
13
14
16
17
19
20
end
```

In [91]:
```
ava=5
x=int(input('How many chocolates do u want?'))

i=1
while i<=x:
    print('chocolate')
    i+=1
```

```
chocolate
chocolate
chocolate
```

In [ ]:

In [ ]:

In [ ]:

```

```